Single Photon Quantum Optics

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Abstract

This experiment investigates the quantum nature of light through single-photon interference and coincidence measurements. Spontaneous parametric down-conversion (SPDC) generates pairs of entangled photons by splitting a single high-energy photon into two lower-energy photons. These photon pairs are then directed through a beam splitter and a polarization interferometer. The detection setup consists of either two detectors monitoring the idler photons for coincidences or three detectors: one measuring the signal photon and two measuring the idler photon. This allows us to determine whether light exhibits wave-like, particle-like, or both behaviors. The three-detector measurements show antibunching and anticorrelation in the intensity correlation function, supporting the quantum theory of light as indivisible photons. In contrast, two-detector measurements align with classical behavior for a coherent wave or classical light. Together, these results provide experimental evidence for wave-particle duality and highlight the importance of measurement in quantum optical systems.

1 Introduction

This project investigates the quantum nature of light through single-photon experiments. We sought to address a fundamental question: does light behave as a particle, a wave, or both? We were able to determine how individual photons behave when incident on a beam splitter and how measurement influences their observed properties.

The nature of quantum theory implies that a particle upon a beamsplitter may be registered at exactly one detector following it. This defines the particle as an indivisible entity with a $\frac{1}{2}$ probability of taking each negatively anticorrelated path. Classical fields, on the other hand, treat it as a classical wave under a function of space and time, where we can predict some coincidences at both ends as light propogates. In this case, with single-photon light incident on a beamsplitter, the electric fields in the upper and lower paths have equal magnitudes, but differ by a multiplicative phase; classical theory then predicts a positive correlation between the two detectors.

This is the experimental concept for the Grangier-Roger-Aspect experiment, which examined correlations between photodetections at a beamsplitter. Our modern interpretation provides evidence for both the quantum and classical nature of light, underscoring its significance for understanding light's fundamental properties.

2 Experimental Goals

Using coincidence techniques, we experimentally showed the wave-particle duality of light. Under classical theory, where light acts as a wave, we may assume that light split by a 50/50 beamsplitter can be detected on both paths. However, quantum theory suggests that light acts solely as a particle, where when incident upon a beamsplitter, only one detector will "fire".

3 Setup

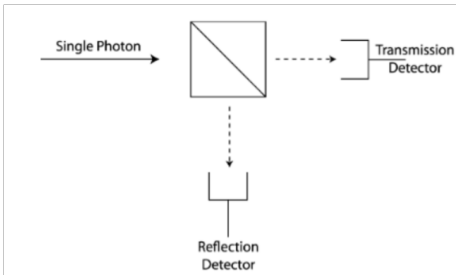


Figure 1: This image shows the lab's conceptual scheme assuming a reliable single-photon source. This stream of photons is incident on a 50/50 beamsplitting cube. If the photons were completely indivisible, then only one detector would record a "firing", so we should find an even split between reflection and transmission detections–but never a simultaneous count.

Our experiment uses a coherent light source (a 405-nanometer, 50-mW violet laser) that emits waves with a constant phase difference. The laser beam first passes through a half-wave plate and is then directed onto a pair of barium borate (BBO) crystals arranged for type-1 spontaneous parametric down-conversion (SPDC). In this nonlinear process, each incoming "pump" photon is instantaneously converted into two lower-frequency entangled photons. The BBO crystals are cut along their optic axes and oriented at 90 degrees to each other, resulting in photon pairs with orthogonal polarizations. After leaving the crystals, one photon, called the "signal," is routed toward the gate detectors (A, A'), while the other, the "idler," takes a separate path toward the final transmission and reflection detectors (B, B'). As they propagate, both signal and idler photons remain entangled in energy, direction, and polarization.

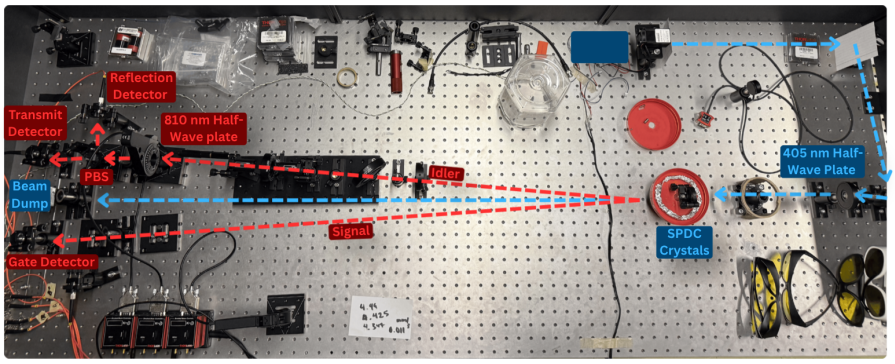


Figure 2: Total setup for the Grangier experiment

The entangled scattered idler photons are sent towards an 805 nm half-wave plate, which can again be optimized for transmission or reflection detections by tuning it towards a given polarization angle–ensuring our PBS is acting like a true 50/50 beamsplitter cube. At an even optimization, the idler photons are split by a beamsplitter cube between our final two detectors. We can use this detection correlation with our gate detector to find the correlations of our entangled pairs.

4 Theory

4.1 Classical Theory

Classical theory defines the electromagnetic wave by Maxwell's equations:

$$g^{(2)}_{T,R}(\tau) = \frac{\langle I_T(t+\tau)\, I_R(t)\rangle}{\langle I_T(t+\tau)\rangle\langle I_R(t)\rangle} \tag{1}$$

Showing that the correlations between the intensities of transmitted (I_T) and reflected (I_R) beams are given by the degree of second-order coherence $g^{(2)}_{T,R}(\tau)$. As an ensemble of coherence detections, we evaluate the normalized intensity correlation function at a time delay of $\tau = 0$.

If we redefine the transmission and reflection coefficients T and R to be related to our initial beam intensity rather than scattered light intensities, we have $I_T = TI_0$ and $I_R = RI_0$, where I_0 is the generalized incident intensity. This gives:

$$g^{(2)}_{T,R}(\tau) = \frac{\langle (TI_0(t+\tau))(RI_0(t))\rangle}{\langle TI_0(t+\tau)\rangle\langle RI_0(t)\rangle}$$

$$g^{(2)}_{T,R}(\tau) = \frac{\langle I_0(t+\tau)I_0(t)\rangle}{\langle I_0(t+\tau)\rangle\langle I_0(t)\rangle},$$

$$g^{(2)}_{T,R}(0) = \frac{\langle [I_0(t)]^2 \rangle}{\langle I_0(t) \rangle^2}$$

The Cauchy-Schwarz inequality then finds that:

$$g^{(2)}_{T,R}(0) = g^{(2)}(0) \geq 1$$

Until now, we have assumed a perfectly stable input field; in this scenario, photons follow predictable paths. When considering second-order coherence (a measure of the correlation between intensities at two points in space or time), fluctuating fields produce values greater than 1, indicating a chaotic source. In such a case, photons undergo collisional or Doppler broadening and take random paths. By contrast, in our case, the source is pseudothermal, meaning it behaves like thermal light but is generated by a coherent source, or it is simply a coherent source. Here, the light waves remain fixed in time as they propagate, so wave interference occurs constructively or destructively in a predictable way. Consequently, the measured intensity correlation function, which shows how intensities at two points relate, should classically be 1 for a coherent source. If we instead assume no coherence between detectors, as in the quantum case, the numerator vanishes, and the correlation function approaches 0.

4.2 Interpreting single-photon behavior

In calculations, the assumption is made that a steady stream of photons exists. This suggests that constructing a light source by attenuating a laser beam so that only one photon is present in the vicinity of the beamsplitter at any given time is necessary. This enables direct photon-count measurement and is essential for fully testing the outlined hypothesis. The semiclassical theory simplifies the lab setup by assuming the electromagnetic field is classical, while the photodetector acts quantum mechanically.

The conversion of continuous electromagnetic radiation into discrete photons is then a random process, and the probability of obtaining a single photocount from any given detector within Δt, being the time window in which our detectors can record a photon, is proportional to the average intensity of the field striking said detector. This is shown by:

$$P_T(\Delta)t = \eta_T \langle I_T(t) \rangle \frac{\Delta t}{\tau_T}$$

Where η is the constant efficiency coefficient of detector T. The joint probability of detecting both a transmission and a reflection count is given by:

$$P_{TR}(\tau)\, \Delta t^2 = \eta_T \eta_R \, \langle I_T(t+\tau)\, I_R(t) \rangle \, \Delta t^2 \tag{2}$$

With these two equations, we can then measure the second-order coherence in terms of probability rather than intensity:

$$g^{(2)}_{T,R}(\tau) = \frac{\eta_T \eta_R \, \langle I_T(t+\tau)\, I_R(t) \rangle \, \Delta t^2}{(\eta_T \langle I_T(t) \rangle \Delta t)\, (\eta_R \langle I_R(t) \rangle \Delta t)} = \frac{P_{TR}(\tau)}{P_T P_R}$$

Now we can mathematically prove that classical theory gives the bound $g^{(2)}(0) = 1$, meaning the light propagates and hits both the T and R detectors completely uncorrelated. This occurs when the input field is a perfectly stable wave. For more chaotic or thermal light sources, $g^{(2)}(0)$ increases beyond this and may even approach $g^{(2)}(0) = 2$.

Since $g^{(2)}(0) \geq 1$ under classical theory, the T and R detection events cannot exhibit antibunching. Instead, the fluctuations in intensity are positively correlated, meaning that increases in intensity at one detector are accompanied by increases at the other.

4.3 Quantum Theory

According to quantum theory, when the electromagnetic field is quantized, the transmitted and reflected intensities are described by second-order coherence $g^{(2)}_{T,R}(\tau)$, where the intensities are treated as quantum operators rather than classical waves:

$$g^{(2)}_{T,R}(0) = \frac{\left\langle : \hat{I}_T \hat{I}_R : \right\rangle}{\left\langle \hat{I}_T \right\rangle \left\langle \hat{I}_R \right\rangle}$$

The colons indicate normal ordering of the creation and annihilation operators corresponding to the electric field. The intensity operator is proportional to the photon number operator for the field, $\hat{n} = \hat{a}^\dagger \hat{a}$, so in substitution we can find:

$$g^{(2)}_{T,R}(0) = \frac{\langle : \hat{n}_T \hat{n}_R : \rangle}{\langle \hat{n}_T \rangle \langle \hat{n}_R \rangle} = \frac{\left\langle \hat{a}^\dagger_T \hat{a}_T \hat{a}^\dagger_R \hat{a}_R \right\rangle}{\left\langle \hat{a}^\dagger_T \hat{a}_T \right\rangle \left\langle \hat{a}^\dagger_R \hat{a}_R \right\rangle}$$

The averages here are given by quantum-mechanical expectation values computed from the field states at the detectors. The field operators are written in terms of the input field operator $\hat{a}_I$ and the vacuum field operator $\hat{a}_V$. For a 50/50 beamsplitter, this means:

$$\hat{a}_T = \frac{1}{\sqrt{2}} \left(\hat{a}_I + \hat{a}_V \right), \qquad \hat{a}_R = \frac{1}{\sqrt{2}} \left(\hat{a}_I + \hat{a}_V \right)$$

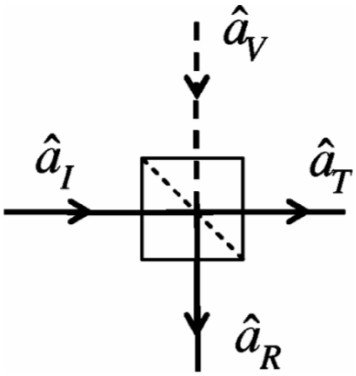


Figure 3: Shows the field operators corresponding to our detection ports

And substituting this back into our original equation, we can find our intensity correlation equation to be the following:

$$g^{(2)}_{T,R}(0) = \frac{\langle \hat{n}_I(\hat{n}_I - 1)\rangle}{\langle \hat{n}_I\rangle^2} = g^{(2)}_{I,I}(0) = g^{(2)}(0)$$

The quantum mechanical equivalent of a stable classical wave is the coherent state $|\alpha\rangle$, which is the eigenstate of the annihilation operator:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle.$$

Evaluating $g^{(2)}(0)$ for this input field yields $g^{(2)}(0) = 1$ for a coherent state. From a chaotic light source, we obtain $g^{(2)}(0) = 2$, where photons tend to hit the detectors in "bunches," leading to a positive correlation in the output fields.

This shows that our quantum theory equation agrees with classical theory in the appropriate field states. However, in the case of single photon interference, the eigenstate of the number operator with eigenvalue 1:

$$\hat{n}|1\rangle = 1|1\rangle$$

Evaluating gives $g^{(2)}(0) = 0$, violating the classical inequality $g^{(2)}(0) \geq 1$.

This shows that photons do not stay together and instead tend to arrive one at a time, leading to anticorrelations in the photocounts. The probability can then be described by a Poisson distribution:

$$P(n, \Delta t) = \frac{\langle n\rangle^n}{n!} e^{-\langle n\rangle}$$

$\langle n\rangle$ represents the average number of photons recorded in Δt. This equation can be used to show that the ratio of probabilities of detecting 2 photons in a single time interval is given by:

$$\frac{P(2,\Delta t)}{P(1,\Delta t)} = \frac{\frac{\langle n\rangle^2}{2!}e^{-\langle n\rangle}}{\frac{\langle n\rangle^1}{1!}e^{-\langle n\rangle}} = \frac{\langle n\rangle}{2}$$

Here, Δt is the time taken for the light to travel through the vicinity of the beamsplitter, so if the laser is attenuated such that $\langle n\rangle \ll 1$, there is a high probability of single-photon interference. However, since calibration and available equipment make this task so challenging, the normalized average counts are used as the final recorded data.

5 Calibration

Our experiment began by calibrating the 405 nm half-wave plate (HWP), as shown in Figure 2, by collecting data at slow-moving angles to determine where either the transmission detector (B, which measures photons passing through the beamsplitter) or the reflection detector (B', which measures photons reflected off the beamsplitter) is optimized for the most photon counts.

We started at 0° and increased by 5° increments. We found that transmission detections are minimized at 23.24°, and the reflection detector is similarly optimized at the same point.

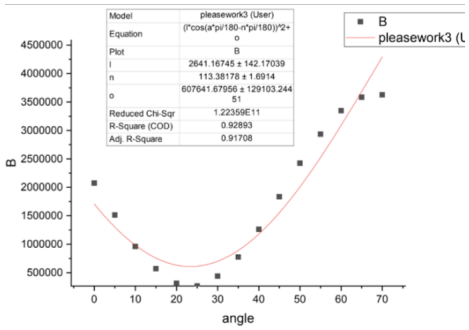


Figure 4: Calibration curve fit for the 405 HWP for the transmission detector

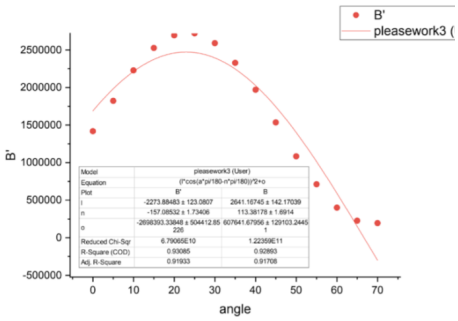


Figure 5: Calibration curve fit for the 405 HWP for the reflected detector

We kept our optimization point at 23.24°. For the remainder of this experiment, this value remained unchanged.

Unlike the 405 HWP, the 805 HWP does not need to be adjusted manually and offers far greater precision (0.05° rather than 0.5°). We followed a similar procedure and our final data is shown below:

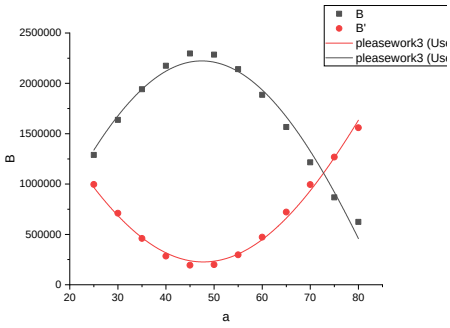


Figure 6: Calibration curve fit for the 805 HWP for the transmitted detector

The coincidence program showed that B transmission is optimized at 47.2°.

6 Experimental Processes

6.1 Probability Definition

From our intensity equations, we have our intensity correlation interpretation related to probability by:

$$g^{(2)}_{T,R}(\tau) = \frac{P_{TR}(\tau)}{P_T P_R}$$

However, this once again introduces the fundamental issue that we cannot measure either probability or intensity. Our detections measure single photocounts, so we must redefine this equation once again using the measured detections, N.

$$N_T(\tau) = P_{(T \text{ in } \Delta t)} \frac{\tau_T}{\Delta t}$$

This equation shows our detection count N_T over the period τ to be the probability of a photon being present at T over τ, multiplied by the time period τ divided by our detection interval.

We can repeat this process for other detectors and substitute them into our intensity correlation function. The full substitution process is shown below.

$$P_{T \text{ in } \Delta t} = N_T(\tau)\frac{\Delta t}{\tau}, \qquad P_{R \text{ in } \Delta t} = N_R(\tau)\frac{\Delta t}{\tau}, \qquad P_{TR \text{ in } \Delta t} = N_{TR}(\tau)\frac{\Delta t}{\tau}$$

$$g^{(2)}_{T,R}(0) = \frac{P_{TR \text{ in } \Delta t}}{P_{T \text{ in } \Delta t}\, P_{R \text{ in } \Delta t}} = \frac{N_{TR}(\tau)\frac{\Delta t}{\tau}}{\left(N_T(\tau)\frac{\Delta t}{\tau}\right)\left(N_R(\tau)\frac{\Delta t}{\tau}\right)} = \frac{N_{TR}(\tau)\,\tau}{N_T(\tau)\,N_R(\tau)\,\Delta t}$$

Now, with our two-detector measurements, we are left with a coefficient $\tau/\Delta t$, which must be determined. However, if we repeat the same procedure for the three-detector measurements, we find that the intensity correlation can be expressed entirely in terms of our photocounts:

$$g^{(2)}(0) = \frac{N_{GTR}(\tau)\left(\frac{\tau}{\Delta t}\right)^3 N_G(\tau)\left(\frac{\tau}{\Delta t}\right)}{N_{GT}(\tau)\left(\frac{\tau}{\Delta t}\right)^2 N_{GR}(\tau)\left(\frac{\tau}{\Delta t}\right)^2} = \frac{N_{GTR}(\tau)\,N_G(\tau)}{N_{GT}(\tau)\,N_{GR}(\tau)}$$

Without the coefficient $\tau/\Delta t$, we can record data from three detector measurements and later calibrate our measurements based on the equipment's variable Δt.

6.2 $g^{(2)}(0)$ Measurements with Three Detectors

Our next step was then to take the three-detector (G, T, R or A, B, B') measurements using heralded photons. In this context, a photon detected in the G detector heralds the presence of a photon in the TR arm, so we only consider TR coincidences when G has recorded a detection. This use of heralded photons allows us to condition our correlation measurements and is mathematically represented by the photocounts N_{GR} rather than N_R, as shown in section 6.1.

We conducted a 20-minute uninterrupted test run to obtain an accurate average $g^{(2)}(0)$ measurement.

Our final result was $g^{(2)}(0) = 0.175913$, aligning closely with the quantum expectation, where the coherence function approaches zero.

6.3 Calibration of the Detection Window Δt

To calibrate the coincidence time window, we removed short-wavelength light and ensured that photons reached the detectors independently. This was achieved by scattering blue laser light with a notecard, resulting in approximately 10^5 detections for each G, T, and R detector, with the light now uncorrelated, in preparation for solving our two-detector measurements.

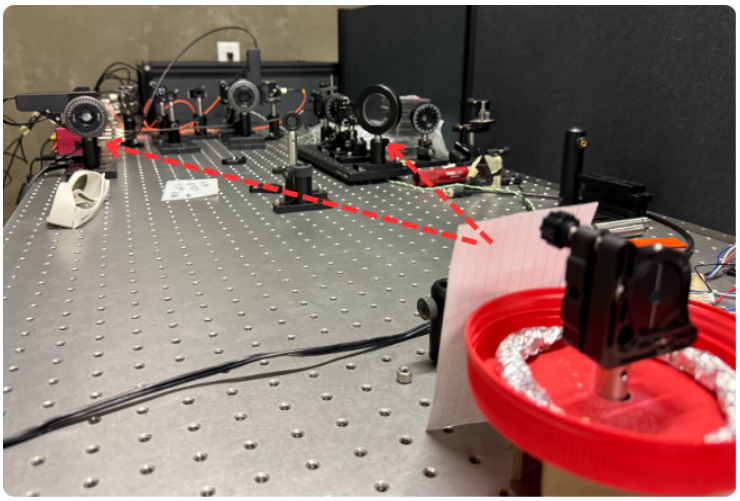


Figure 7: Showing the experimental setup of our Δt calibration

Once this setup was complete, we conducted a 20-minute data run with 10-second intervals. Assuming the uncorrelated light yields $g^{(2)}(0) = 1$, we calculated the value of Δt as follows:

$$\frac{N_{TR}}{N_T N_R}\left(\frac{\tau}{\Delta t}\right) = 1$$

$$\frac{N_{TR}}{N_T N_R}\tau = \Delta t$$

Where τ is the 10-second measurement interval, and N represents the average photocounts recorded within each interval.

*number of point	60		Average B' =	5119693.767
*time per step (s	10		Average BB' =	11063.61667
*BB' coincidence	7.390000 +/- 0.030000		Average B =	1565952.067
*average g(2)	1.865082			
*standard dev. of	0.019288		delta t =	0.0000000137998597
*expected g(2)	1			

Figure 8: Presents the final dataset obtained from the blue-light scattering calibration measurement.

We determined $\Delta t = 13.8 \times 10^{-9}$ s.

6.4 $g^{(2)}(0)$ Measurements with Two Detectors

For two-detector measurements at the T and R detectors, we maintained the standard program settings and subsequently recalculated $g^{(2)}(0)$ using our calibrated value of Δt:

*average g(2)	1.819787			
*standard dev. of	0.103638		Actual g(2) =	0.9931707416
*expected g(2)	1		Percent Error =	0.1041234358

Figure 9: Presents the final dataset obtained from the two-detector measurement of $g^{(2)}(0)$

Our experimental data for the two-detector measurements showed $g^{(2)}(0) = 0.99317$, which closely follows the Poisson distribution predicted by classical light theory.

6.5 Error Propogation

To account for systematic errors, we calculated the standard deviation for each N for relative uncertainty and determined the corresponding error propagation in the overall intensity correlation function:

For three detector measurements:

$$\sqrt{\left(\frac{\sigma_g(2)}{\langle g(2)\rangle}\right)^2} = 0.34413 \approx 34\%$$

$$\sigma\bar{g} = \frac{\sigma_g}{\sqrt{n}} = 0.002745$$

Our result for three detectors of $g^{(2)}(0) = 0.175913$ is then 300.16 deviations below 1, which is the classical requirement. This further shows that a strong deviation proves that our data does not align with classical theory.

For the two-detector measurements, we first needed to determine the error in Δt, so we used the relation $\Delta t = \tau \frac{N_{TR}}{N_T N_R}$ to find the percent error to be 0.099227. From there, we incorporated this into our overall error propagation to obtain:

$$\sqrt{\left(\frac{\sigma_g(2)}{\langle g(2)\rangle}\right)^2 + \left(\frac{\sigma_t}{\langle t\rangle}\right)^2} = 0.1041234 \approx 10\%$$

$$\sigma_{\bar{g}} = \frac{\sigma_g}{\sqrt{n}} = 0.013463$$

Our result of $g^{(2)}(0) = 0.99317$ is 0.507 deviations from the mean and 0.066 standard deviations away from the classical g(0)=1. This demonstrates a strong alignment with classical theory.

6.6 Single-Photon Interference

Using our entangled-state source, we added a polarization interferometer along the idler path composed of two BBO crystals cut for type-I downconversion, stacked back-to-back and oriented at 90° relative to one another. The first crystal converts vertically polarized pump photons into horizontally polarized signal and idler photons, while the second crystal converts horizontally polarized pump photons into vertically polarized pairs. This configuration allows us to vary the pump polarization and control the relative phase between the horizontal and vertical polarization components. A half-wave plate ($\lambda/2$) placed between the crystals is used to adjust the pump polarization.

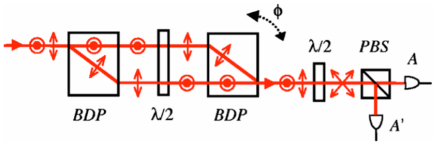


Figure 10: The polarization interferometer controlling the relative phase and polarization of the photons passing through

Our original source produces photon pairs in the entangled state:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|H\rangle_s |H\rangle_i + |V\rangle_s |V\rangle_i \right),$$

where H and V denote horizontal and vertical polarization.

There is a 50/50 probability of generating either polarization pair; however, the state also implies strong correlations: if the idler photon is measured to be horizontally polarized, the signal photon must also be horizontally polarized, and similarly for vertical polarization. This expression represents the entangled quantum mechanical state of the photon pair.

In the experiment, we rotated the half-wave plate by 45°. This adjustment caused the two polarization components (the directions in which light waves vibrate) to overlap both spatially (in position) and temporally (in time). The beams recombine upon exiting the interferometer. Rotating the second half-wave plate alters the polarizations again. This allows us to observe how the photons interfere at the beamsplitter.

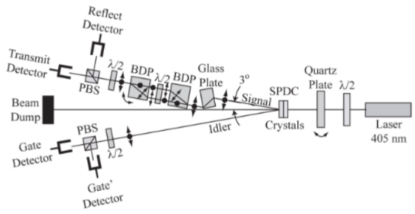


Figure 11: The full experimental setup, including the signal and idler photon paths

As single photons pass through the interferometer, interference fringes appear only when the which-path information of the idler photons is unkown. The fringes are shown below in figure 12 and 13.

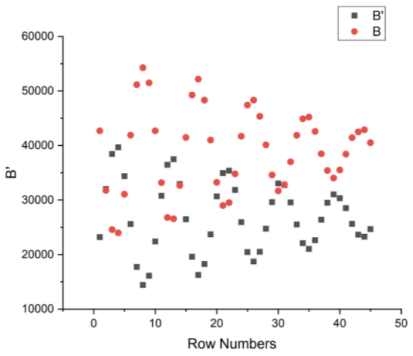


Figure 12: This figure shows the interference fringes that appear when the 810 nm half-wave plate is vertically optimized

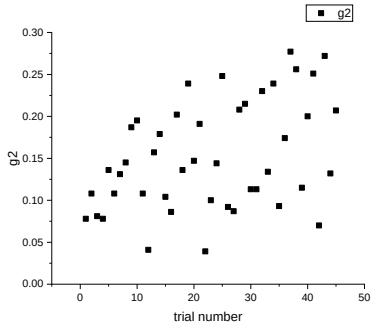


Figure 13: This figure shows g(0) for TR counts as recorded in Figure 12. This shows low-intensity correlation values indicating particle-like behavior, despite wave-like patterns in the detector counts

Although our normalized intensity correlation values shown in Figure 13 are near zero, indicating quantum particle behavior, we can still see classical wave-like interference in Figure 12 between the transmission and gate detectors.

7 Results

This project explores the quantum nature of light using single-photon experiments. We examine whether light behaves as a particle, a wave, or both by observing photon behavior at a beam splitter and examining its varied outcomes.

We used spontaneous parametric down-conversion (SPDC) to generate pairs of entangled photons, which were directed through a beam splitter and detected using coincidence counters. Careful calibration of the half-wave plates and the implementation of a precise coincidence time window enabled accurate measurement of photon correlations. Experiments were conducted with two- and three-detector configurations to compare the normalized intensity correlation predicted by classical and quantum theories.

In the three-detector configuration, we observed a strong anticorrelation in the normalized intensity ($g^{(2)}(0) \approx 0.18$), which confirms the quantum behavior of single photons—where a photon is transmitted or reflected, but not both. In contrast, the two-detector measurement yielded $g^{(2)}(0) \approx 1$, consistent with the classical light behavior.

By adding a polarization interferometer along the idler photon path, we controlled the relative phase and polarization of photons passing through. We then observed interference fringes despite our normalized intensity correlation approaching zero, indicating particle behavior.

8 References

References

[1] P. Grangier, G. Roger, and A. Aspect, *Experimental Evidence for a Photon Anticorrelation Effect on a Beam Splitter: A New Light on Single-Photon Interferences*, Journal de Physique Lettres **46**, L-247–L-252 (1985). doi:10.1051/jphyslet:019850046012024700

[2] P. Grangier, G. Roger, and A. Aspect, *Experimental Evidence for a Photon Anticorrelation Effect on a Beam Splitter*, Europhysics Letters (1986).

[3] J. J. Thorn, M. S. Neel, V. W. Donato, G. S. Bergreen, R. E. Davies, and M. Beck, *Observing the Quantum Behavior of Light in an Undergraduate Laboratory*, American Journal of Physics **72**(9), 1210–1219 (2004). doi:10.1119/1.1761064

[4] A. Gogo, W. D. Snyder, and M. Beck, *Comparing Quantum and Classical Correlations in a Quantum Eraser*, Physical Review A **71**, 052103 (2005). doi:10.1103/PhysRevA.71.052103

[5] *Article from Physical Review A*, Physical Review A **83**(3), 033837 (2011). doi:10.1103/PhysRevA.83.033837

[6] P. Grangier, *Single Photon Interference Lecture Notes*, University of Illinois Physics 513 course material (1986).
`https://courses.physics.illinois.edu/phys513/sp2016/reading/week1/GrangierSinglePhoton1986.pdf`

[7] *GRACOM Notes on Quantum Optics*, University lecture notes/reference document.
`https://www.math.umb.edu/~sp/gracom.pdf`